Evidence-backed, interactive health-insurance coverage simulator

TRACK 2 · AI-POWERED FINANCIAL JOURNEYS

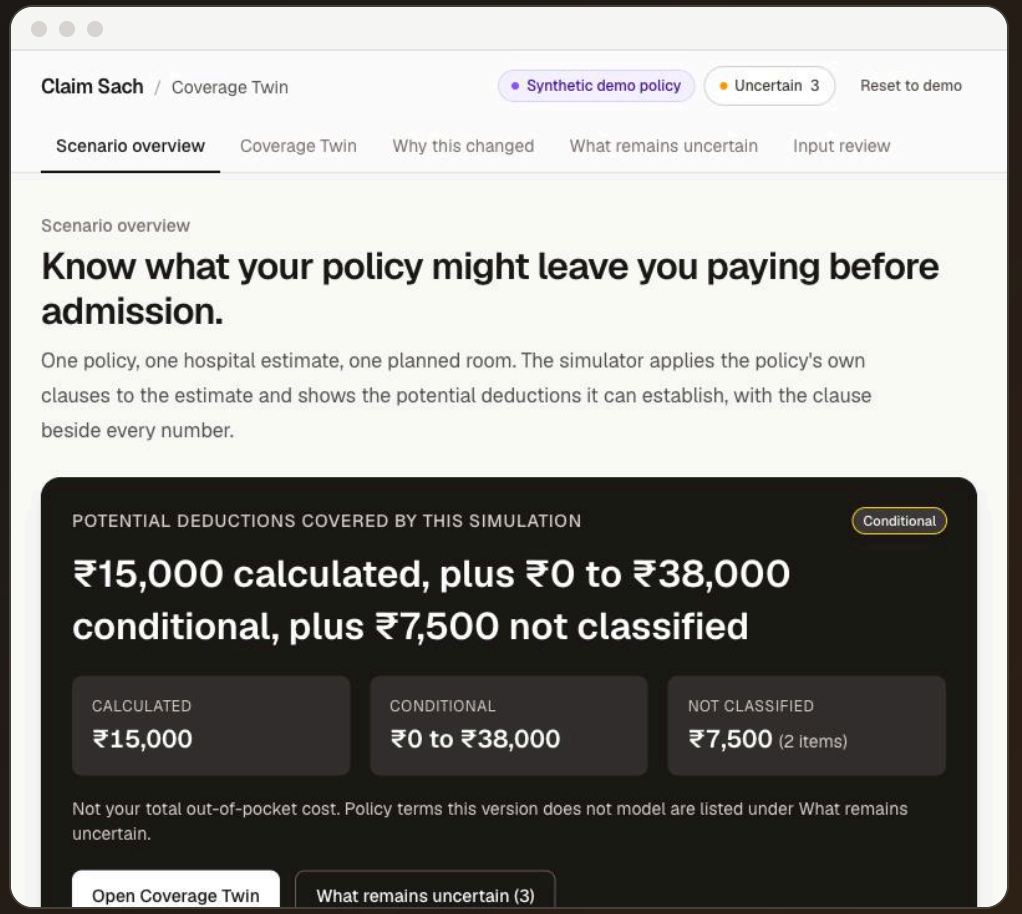
Claim Sach

Coverage Twin

Know what your policy might leave you paying before admission.

Pushkar Mondal · Paytm Build for India AI Hackathon, Bengaluru · October 2026

Working prototype · deterministic engine · 180 tests · a real IRDAI-hosted wording encoded and measured · runs offline

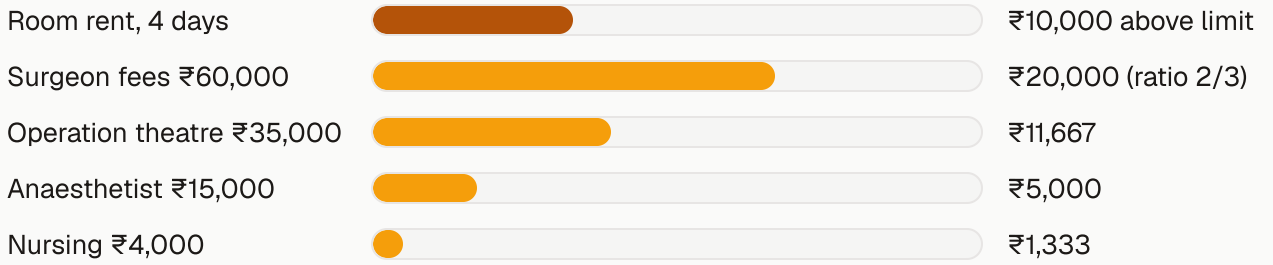


PROBLEM STATEMENT

The policy's cost shows up at the hospital

Room-rent limits, proportionate deductions, sub-limits and exclusions surface at admission, not at purchase. The room-rent clause is the sharpest: pick a room above the eligible rate and many policies pay the surgeon, anaesthetist, nursing and operation-theatre charges only in the proportion *eligible ÷ actual rate*.

One choice, four items that are not the room



Worked example on a synthetic policy and estimate, computed by the prototype. Not a market statistic.

Which items are touched depends on the policy's own definition. General explainers stop at the concept; the gap we are testing is applying *your* wording to *your* bill.

₹2,500 a day

the room-rent gap: ₹7,500 chosen against ₹5,000 eligible

₹48,000

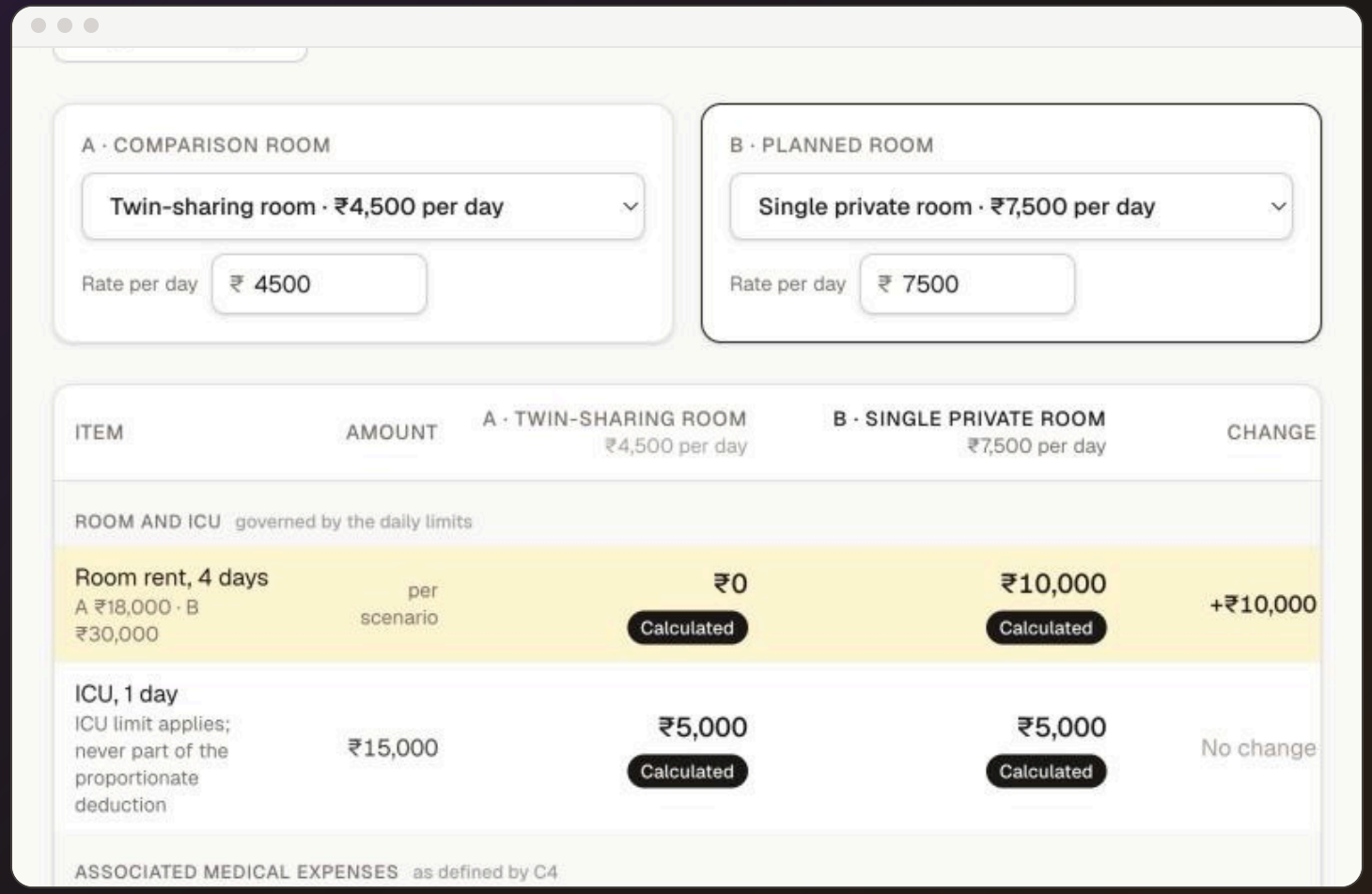
potential deductions from that one choice; ₹38,000 of it on items that are not the room

4 of 11 lines

affected under the policy's definition; 3 excluded by it; 2 the wording never classifies

THE MOMENT

Change your room. See the difference.



1. **Two rooms, one policy, one estimate**
   Twin-sharing at ₹4,500 against single private at ₹7,500. The amber rows are what changed, and only those.

2. **Every amount opens its evidence**
   Clause and page, the inputs with where they came from, the formula, the result. Tampered traces fail validation.

3. **Ranges, not guesses**
   Until the hospital confirms differential billing, the surgeon's deduction is ₹0 to ₹20,000. Confirm it on Input review and it becomes ₹20,000, with "confirmed by you" on the trace.

4. **Unknown is never zero**
   Two lines the wording never classifies stay out of every total, and the panel says who can settle them.

PROPOSED SOLUTION

Coverage Twin: your wording, your bill, live

1. **Policy documents**
   Wording and schedule, synthetic or real, labelled which.
2. **Structured rules**
   Each rule cites its clause and quotes it in full.
3. **Scenario simulation**
   Room, rate, days, the hospital's own line items.
4. **Itemised potential deductions**
   Room excess, ICU excess, proportion, co-pay.
5. **Source clause**
   Page-cited, verbatim, beside every number.
6. **Explicit uncertainty**
   What is open, what is at stake, who can settle it.

0:00 · Overview

₹15,000 calculated, ₹0 to ₹38,000 conditional, ₹7,500 not classified. Not a total, and it says so.

0:25 · Twin

Change the room; the itemised breakdown updates at once with the cause rule per row.

1:35 · Uncertain

Ask the hospital, ask the insurer, resolve now; plus what this version does not model at all.

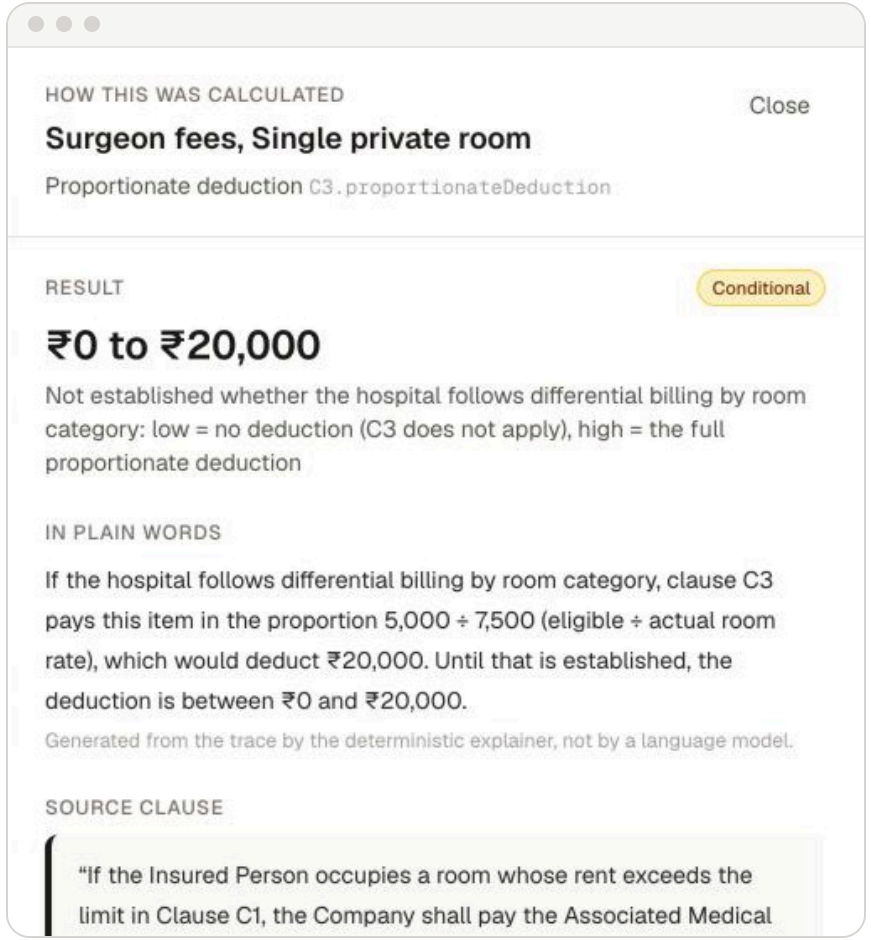
3:00 · Real wording

Switch to Arogya Sanjeevani and watch the same engine refuse to guess where the wording is silent.

Not a chatbot. Not a payout guarantee. A scenario estimate of the deductions it covers, with its unknowns on screen.

PROOF, NOT PROMISES

A real wording, quoted verbatim, with the conflict it contains



INSURER-ISSUED · IRDAI-HOSTED · PARTIALLY ENCODED

Arogya Sanjeevani Policy, The New India Assurance Co. Ltd.

UIN NIAHLIP20165V011920 · downloaded from irdai.gov.in on 24 October 2026 · 31 pages · ten clauses encoded with page numbers · SHA-256 4e261c78…f542d44 shown on screen

What the engine applies

Room 2% of SI up to ₹5,000 a day; ICU 5% up to ₹10,000; the proportion of all other expenses except medicines when either rate is exceeded; a 5% co-payment.

What it refuses to guess

This 2019-20 wording reduces implants and diagnostics; IRDAI's June 2020 norms later barred that. The engine shows those items as a range and says "requires insurer confirmation".

The ICU surprise

A ₹15,000 ICU day alone triggers the proportion under this wording, even in a twin-sharing room. When room and ICU exceed in different ratios, the engine gives bounds, not a number.

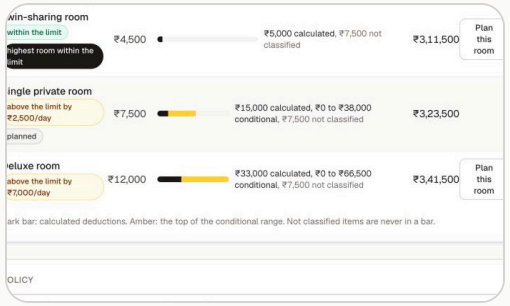
Measured

21 of 21 hand-computed amounts exact; 3 of 3 refusal cases right; every trace validated against the verbatim clause text.

Partial encoding, labelled so everywhere. The schedule used with it is synthetic; real schedules are personal and never bundled.

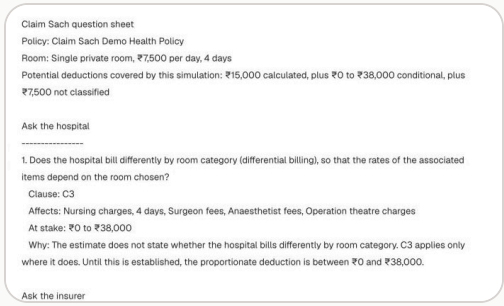
WHAT SHIPS TODAY

Four tools around one engine



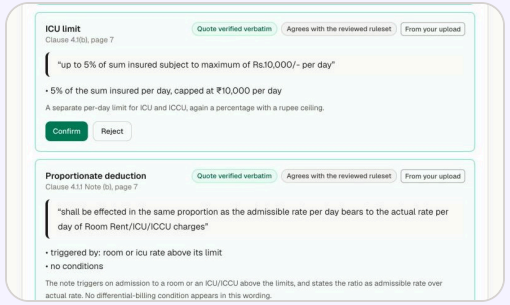
Room ladder

Every room on one view with the deductions at each; the highest room within the limit is marked. One click plans it.



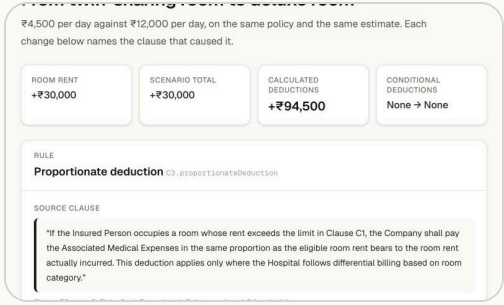
Question sheet

Every open uncertainty becomes a question for the hospital or the insurer, tied to its clause and the rupees at stake. Copy, print or save.



Ruleset builder

A model proposes rules from a public wording with quotes; the engine verifies each quote verbatim; a person confirms. Cached proposals here; live adapter off.



Why this changed, and the evidence pack

Each change names its clause, the inputs that differ and both formulas. The whole result exports as a Markdown or JSON evidence pack with every trace.

TECHNOLOGY / TECH STACK

Deterministic engine, AI only with human review

Frontend

TypeScript, strict. Next.js 16 (App Router), React 19, Tailwind 4. Six screens on a tested screen model; no business logic in components.

Rules engine

Pure TypeScript, no I/O, no model. Integer paise, exact ratios, one half-up rounding. A trace on every amount: rule, inputs, clause, formula, output, open assumptions.

Two wordings

A synthetic demo policy, and Arogya Sanjeevani (New India, from irdai.gov.in): ten clauses quoted verbatim, page-cited, SHA-256 shown. Partially encoded and labelled so.

AI, in its place

The Ruleset builder: a model proposes rules with quotes; the engine verifies each quote verbatim; a person confirms. Cached in this build; the live adapter is off. Never in the calculation path.

Quality

Vitest, 180 tests, written first. GitHub Actions on every PR. Evaluation harness: six measures over two wordings, results committed and checked for staleness.

Runs anywhere

One command, offline on localhost; fonts bundled; Vercel-ready. Nothing the user enters leaves the browser. No key, no account, no analytics, no database.

DATA FLOW, ALL IN THE BROWSER

```
fixture or upload (JSON, CSV) → ingestion: Zod schema + verbatim evidence check → screen model → rules engine: classify → limits → proportion → co-payment → traces → six screens, question sheet, evidence pack
```

AI seam, off by default: public clause text → model proposals → verbatim quote check → human confirmation → ruleset. No schedule, bill or name ever leaves the browser.

USP

Item-level, clause-cited, honest about unknowns

1. **Every number is clickable**

   Clause and page, inputs with where they came from, the formula, the operation. Tampered traces fail validation.

2. **Unknown is never zero**

   Items the wording does not classify stay out of every total. A conditional amount shows both bounds and the condition.

3. **Change your room, see the difference**

   Two rooms side by side, the rule behind every changed row; the ladder and the question sheet turn it into a decision and a call script.

4. **Proven on a real wording, honestly**

   Arogya Sanjeevani, ten clauses verbatim with a checksum; where the 2019-20 text conflicts with IRDAI's 2020 norms, the engine says so.

What we claim

Claim Sach turns a policy wording into a verifiable hospital-bill simulation: change the room, inspect every modelled deduction, and see what the documents cannot determine.

What we do not claim

Not the first to read clauses; not a payout guarantee; no accuracy figure beyond the measured ones. A bounded review of nine public sources found no tool doing this; we claim we looked, not that nobody does.

IMPACT & BENEFITS

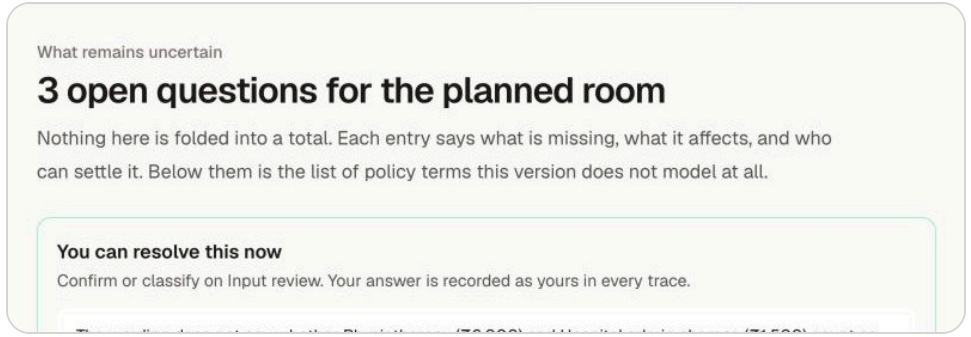
The room decision, made before admission

For the policyholder

The room decision with its deductions visible before admission, not after. Questions for the hospital and the insurer written out: what to ask, and what it is worth.

For a broker's claim-assistance agent

The clause on screen during the call. The uncertainty list is the call script; the evidence pack is the written answer.



MEASURED, NOT CLAIMED

MEASURE	SYNTHETIC	REAL WORDING
Clause retrieval, top-1	11 of 12	10 of 12
Rule extraction, field-level	10 of 10	10 of 12
Calculations, exact paise	9 of 9 · 46 amounts	3 of 3 · 21 amounts
Uncertainty and refusal	6 of 6	3 of 3
Model proposals: quotes verbatim (cached)	n/a	6 of 6
Model proposals: agree with reviewer (cached)	n/a	11 of 13

Measured on the synthetic wording and on ten encoded clauses of one real wording; the model proposals are cached, not a live run. The product shows no accuracy figure anywhere. Figures on this slide are potential deductions covered by the simulation, never total out-of-pocket cost.

BUSINESS MODEL

Where it fits Paytm, and what must be tested

VERIFIED FACTS

- Paytm Insurance Broking Pvt Ltd is a Direct (Life & General) broker: IRDAI Reg. No. 700, renewed 9 Feb 2026, valid to 16 Feb 2029.
- It distributes car, two-wheeler, health, life and term policies from 18+ insurers, including Niva Bupa and Aditya Birla Health. It does not underwrite.
- It already offers claim assistance by phone. Arogya Sanjeevani is IRDAI's standard product on one wording; one verified encoding serves every insurer that sells it.
- FY25 turnover ₹37.1 crore, loss ₹13.2 crore (Form AOC-1), inside a ₹2,593 crore financial-services line in FY26. Insurance is a small line for Paytm today; clarity is a wedge worth testing.

HYPOTHESES, EACH WITH ITS TEST

HYPOTHESIS	TEST
Fewer claim-time support contacts and disputes	Four-week pilot at one claim-assistance desk, against a matched cohort
More eligible-room choices when the consequence is visible	Room-change rate in the simulator, then claims data
Panel insurers co-fund a coverage-clarity tool	Three conversations; one signed pilot letter is the pass
A coverage check lifts conversion or renewal on the policy page	A/B test, if Paytm chooses to run one

Revenue options depend on those tests: a broker-side feature (retention), an insurer-paid integration, or a premium claim-assistance tier. No Paytm integration is claimed or built. Bima Sugam is not live; it belongs in the future tense.

Built, tested, runs offline

```
github.com/Pushkarmondal/paytm-hackathon
npm install && npm run dev
```

180 unit tests · two wordings, one of them real · six evaluation measures · a ruleset builder where a model proposes and a person confirms · a trace on every number.

Next: a second real wording through the builder with the live adapter on; independent review of the evaluation fixtures; a four-week pilot at a broker's claim-assistance desk.

